

NORTH AMERICA WONKA BAR CHOCOLATE

SALES ANALYSIS (2020–2024)

1.PROJECT OVERVIEW:

This project presents a comprehensive business intelligence dashboard designed to analyse and visualize chocolate sales performance across multiple dimensions. Built with a focus on clarity, actionable insights, and professional presentation, the dashboard highlights key metrics and trends that support strategic decision-making.

2.OBJECTIVE:

The analysis has led to six clear objectives. These objectives are derived directly from the findings of this report.

- Identify which chocolate products drive the highest order volumes, helping prioritize production and marketing
 - Compare profitability across regions to highlight strong-performing markets and uncover areas needing strategic improvement.
 - Show shipping mode distribution with Standard Class as the dominant choice. Enable logistics and cost optimization by comparing demand across all modes.
 - Track profit growth trends over multiple years to evaluate long-term performance and measure business sustainability.
 - Compare monthly order values to highlight overall performance trends. Identify peak periods and declining phases to guide sales planning and resource allocation.
 - Pinpoint top-performing cities in terms of order volume, enabling targeted sales campaigns and regional expansion strategies.
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3.DATA SOURCE:

- **Dataset Source:** Maven analyst
 - **Timeline:** 2020-2024
 - **Data Size:** 10,000+ rows and 18 columns
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4.PRE PROCESSING:

- Filled missing values appropriately.
- Removed duplicates
- Standardized text with TRIM (remove spaces) and PROPER (consistent casing).
- Sorted values for structured analysis.
- Validated totals to ensure accuracy after cleaning.

TOOLS USED:

- Excel
 - Power BI (power query)
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5..DATA TRANSFORMATION:

CALENDAR TABLE CREATION

- Established a dedicated **Date table** to support time intelligence function.
- Ensured proper relationships between the Calendar table and the Orders table for consistent filtering.

CORE AGGREGATION MEASURES

- **Total Orders** → Defined as the count of all transactions in the Orders dataset.
 - **Total Sales Amount** → Calculated as the sum of all recorded sales values.
 - **Total Profit** → Derived by aggregating profit values across all orders.
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6.DATA MODELLING:

- Established a structured relationship between the **main Orders table** and the **Calendar table** to enable accurate time-based calculations.
- Ensured one-to-many relationships so that filters applied through the Calendar table propagate consistently across all measures and visuals.

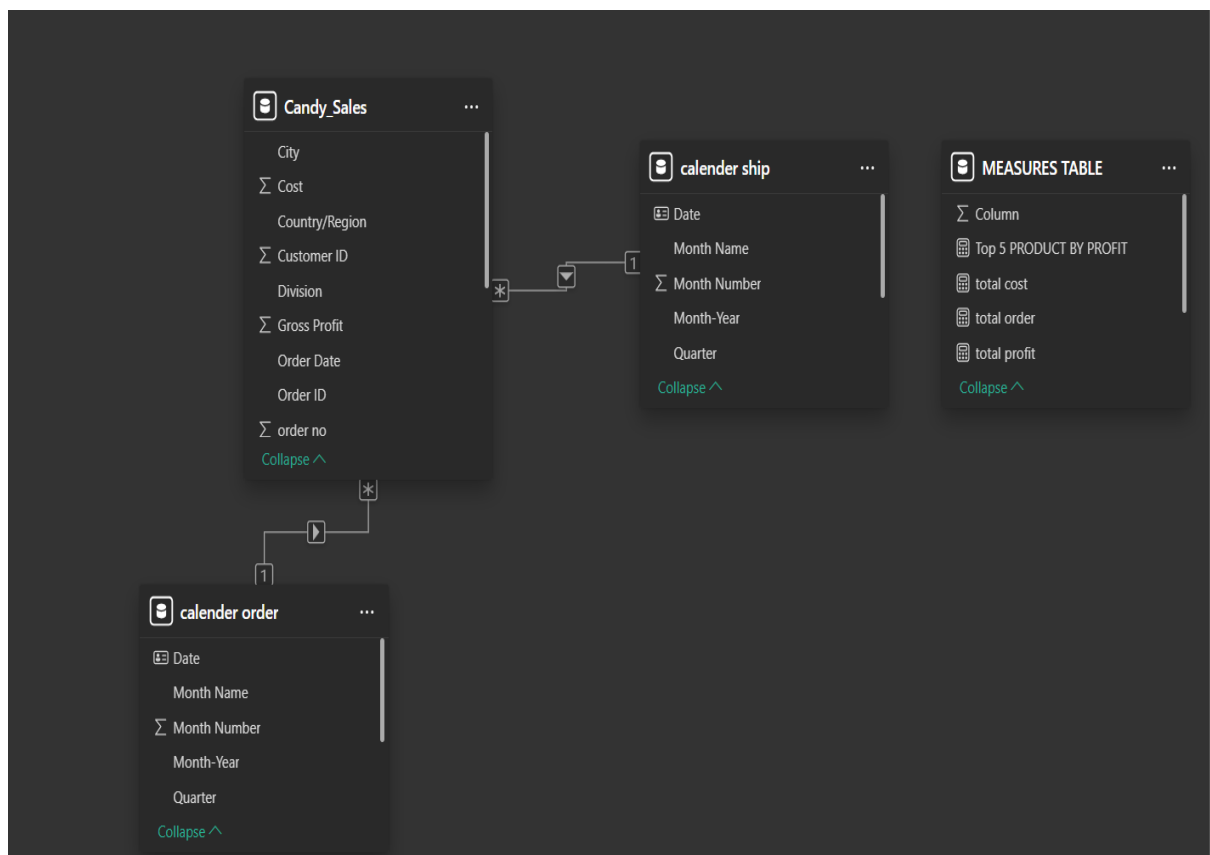


Fig 1: data modelling between the tables

7. EXPLORATORY DATA ANALYSIS (EDA) & VISUALIZATION:

- Used the Calendar table to see yearly and YTD sales/profit growth. Grouped data by product, region, and city to find top performers.
- Checked how orders and profits are spread across regions and products.

Built visuals:

- KPI cards for totals (Orders, Sales, Profit, YTD).
- Bar charts for product/region performance.
- Line chart for yearly profit trend.
- Pie chart for regional order share.

Features Included:

- Filters (Slicers)
- Interactive visuals

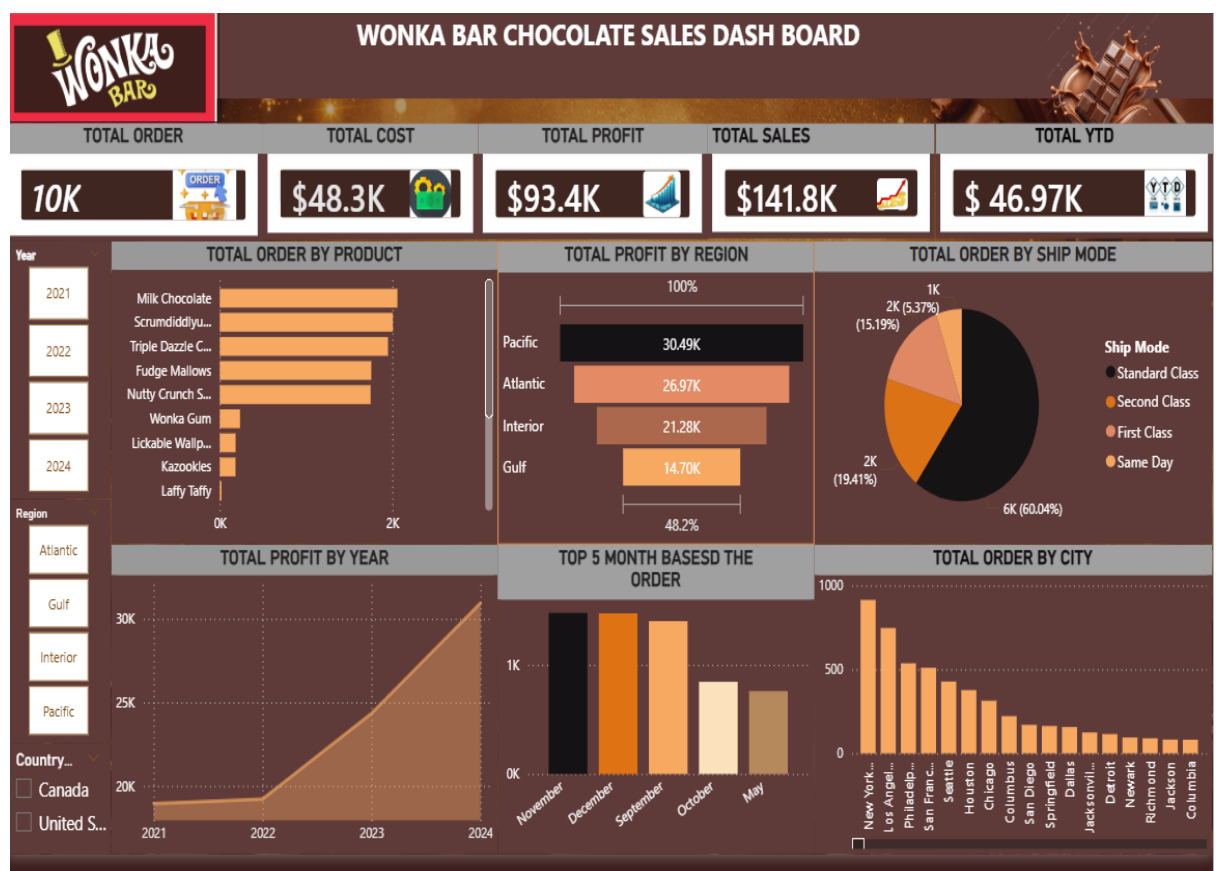


FIG 2: chocolate sales dash board

8. KEY INSIGHTS (SMART + ANALYTICS APPROACH):

8.1: TOTAL ORDER BY PRODUCT:



The following KPIs were identified from the bar chart analysis

1. Descriptive Analytics:

- Milk Chocolate shows strong **Specific** demand dominance, clearly outperforming other products. Mid-range items like Fudge Mallows and Nutty Crunch reflect **Measurable** moderate traction.
- Kahookles and Lickable Wallpaper highlight **Actionable** scope for improvement to achieve **Relevant, Time-bound** growth.

2. Diagnostic Analytics:

- **Milk Chocolate** dominates orders, showing strong consumer preference and reliance on one product. **Fudge Mallows** and **Nutty Crunch** perform moderately, with scope for targeted promotion.
- **Kazookles** and **Lickable Wallpaper** underperform, signalling issues in appeal or positioning.

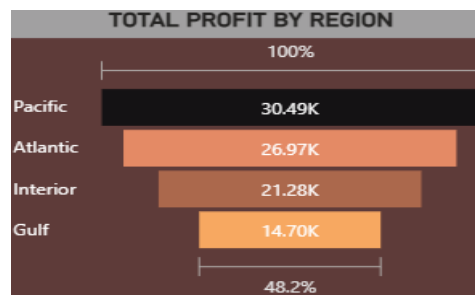
3. Predictive Analytics:

- If the current trajectory continues, profits may reach **36K–40K by 2026**, with top chocolates contributing **60–70%** of total profit.
- Mid-range products will likely sustain moderate traction but need stronger promotion to benefit from the growth momentum. Weaker products will continue lagging unless repositioned, risking missed opportunities in the expanding market.

4. Prescriptive Analytics

- Maintain dominance of **top chocolates** to secure continued profit growth. Strengthen mid-range products through bundling, seasonal campaigns, or targeted promotions to leverage the rising demand curve.
 - Reposition weaker products via pricing, redesign, or niche marketing to align them with the growth trend.
 - Balanced portfolio strategy will reduce reliance on a single product group and sustain long-term profitability in line with the upward trajectory.
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8.2: TOTAL PROFIT BY REGION:



The following KPIs were identified from the funnel chart analysis

1. Descriptive Analytics:

- The Pacific region is the **Specific** leader with the highest profit at 30.49K, while the Atlantic region makes a **Measurable** strong contribution at 26.97K.
- The Interior region shows **Achievable** mid-level traction at 21.28K, leaving scope for growth.
- The Gulf region highlights an **Actionable, Relevant** need for improvement with only 14.70K profit.
- These insights point to a **Time-bound** opportunity to strengthen weaker regions while sustaining top performers.

2. Diagnostic Analytics:

- The **Pacific region** leads with the highest profit, showing strong market strength and customer concentration. The **Atlantic region** follows closely, reflecting competitive performance but slightly less dominance.
- The **Interior region** delivers moderate profitability, suggesting stable yet limited growth potential. The **Gulf region** lags significantly, pointing to weaker demand or operational inefficiencies that need corrective action.

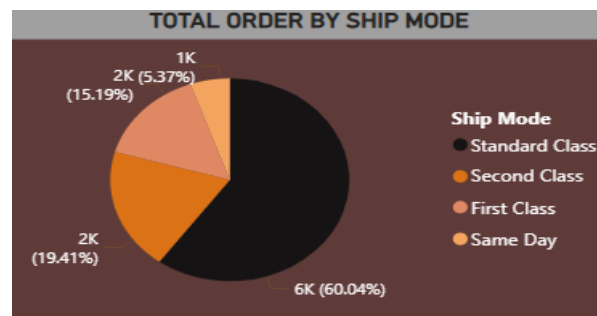
3. Predictive Analytics:

- **Pacific** will remain strong, sustaining profits around **30K+** in the coming years. Atlantic is likely to stay competitive, contributing close to Pacific levels.
- Interior may show steady growth but needs support to expand further. **Gulf** may stay low (~14K) unless distribution improves; with campaigns, it could grow **10–15% yearly**.

4. Prescriptive Analytics:

- **Pacific** should be sustained as the core profit driver through continued investment and customer engagement.
 - **Atlantic** requires competitive positioning and retention strategies to maintain its strong contribution. **Interior** can be strengthened with targeted campaigns and distribution support to unlock growth potential.
 - **Gulf** needs corrective action—improved distribution, pricing, and marketing—to achieve 10–15% yearly growth.
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8.3: TOTAL ORDER BY SHIPMENT MODE:



The following KPIs were identified from the Pie chart analysis

1. Descriptive Analytics:

- The **Standard Class** is the **Specific** leader, dominating with ~60% of total orders (~6K). The **Second Class** provides a **Measurable** share of ~19% (~2K), reflecting moderate adoption.
- The **First Class** holds an **Achievable** 15% (~2K), showing potential for premium growth. The **Same Day** mode highlights an **Actionable, Relevant** need for improvement at ~5% (~1K), requiring focus in the next **Time-bound period**.

2. Diagnostic Analytics:

- The **Standard Class** dominates with ~60% of orders, showing strong customer preference for low-cost delivery. The **Second Class** at ~19% reflects moderate adoption, likely balancing speed and affordability.
- The **First Class** at ~15% indicates niche demand for premium service but limited overall reach. The **Same Day** mode at ~5% highlights weak adoption, pointing to high costs or restricted availability as barriers.

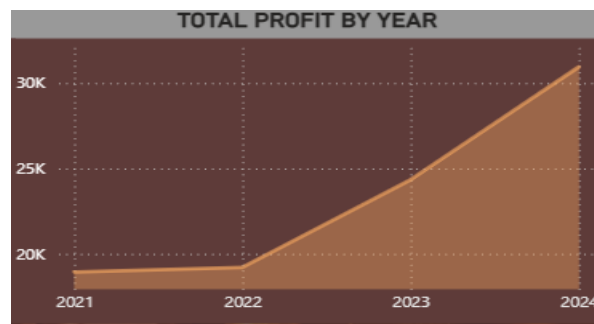
3. Predictive Analytics:

- The **Standard Class** is expected to remain dominant, sustaining ~60% share as customers prefer cost-effective delivery. The **Second Class** is likely to stay steady around ~19%, continuing as a mid-tier option balancing speed and price.
- The **First Class** may show gradual growth beyond ~15% if premium service demand increases. The **Same Day** mode is projected to stay low (~5%) unless pricing and availability improve, where campaigns could drive 10–12% yearly growth.

4. Prescriptive Analytics:

- The **Standard Class** should be maintained as the primary delivery option, ensuring efficiency and cost-effectiveness. The **Second Class** can be optimized with targeted promotions to strengthen its mid-tier adoption.
 - The **First Class** should be positioned as a premium service, leveraging loyalty programs to expand demand. The **Same Day** mode requires corrective action—adjusting pricing, expanding availability, and running campaigns to boost adoption by 10–12% annually.
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8.4: TOTAL PROFIT BY YEAR:



The following KPIs were identified from the Line chart analysis

1. Descriptive Analytics:

- Profits stayed **Specific** at ~20K during 2021–2022, showing stability without major growth. A sharp rise in 2023 made the increase **Measurable**, pushing profits toward 30K+.
- This turning point is **Actionable**, highlighting strategies that can be replicated for sustained success. The upward trend is **Relevant and Time-bound**, signalling momentum to be maintained through 2025–2026.

2. Diagnostic Analytics:

- Profits plateaued at ~20K in 2021–2022, showing dependence on limited product strength and lack of diversification. The sharp surge in 2023 signals successful scaling of top products and improved market traction.
- By 2024, profits exceeded 30K, confirming dominance of high-performing items but exposing reliance on them. Mid-range and weaker products contributed little, highlighting positioning and promotion gaps that hinder balanced growth.

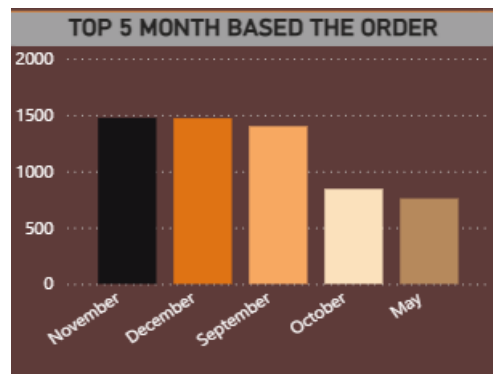
3. Predictive Analytics:

- Profits are projected to continue rising, potentially reaching **36K–40K by 2026** if current momentum holds. Top-performing products will remain the core drivers, contributing around **60–70%** of total profit.
- Mid-range items show moderate traction and will need targeted promotion to capture growth opportunities. Weaker products are likely to lag unless repositioned or redesigned, risking missed potential in the expanding market.

4. Prescriptive Analytics:

- Top-performing products should be sustained as the **core drivers**, ensuring their continued dominance in profit growth. Mid-range items require **targeted promotions, bundling, and seasonal campaigns** to capture rising demand momentum.
 - Weaker products need repositioning through pricing, redesign, or niche marketing to improve their contribution.
 - A balanced portfolio strategy will reduce reliance on a single product group and secure long-term profitability.
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8.5: TOP 5 MONTH BASED THE ORDER



The following KPIs were identified from the column chart analysis

1. Descriptive Analytics:

- Orders stayed **Specific**, with November leading as the peak month, crossing the highest demand levels.
- The difference is **Measurable**, as December and September follow with moderate volumes compared to November.
- This insight is **Actionable**, showing that festive or seasonal demand drives spikes in orders.
- The trend is **Relevant and Time-bound**, highlighting November–December as critical months to sustain growth annually.

2. Diagnostic Analytics:

- **November's** peak highlights strong seasonal or festive demand, showing reliance on one dominant month.
- **December** and **September** follow moderately, suggesting consistent but secondary drivers of order volume.
- **October** and May trail further, indicating weaker consumer engagement or limited promotional impact.
- The imbalance across months signals overdependence on peak seasons, with gaps in demand distribution that need correction.

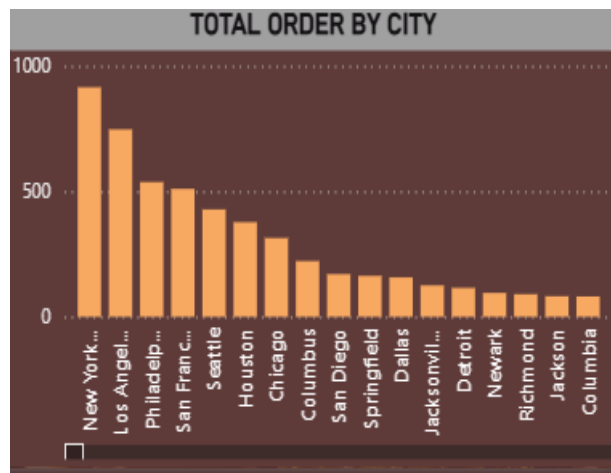
3. Predictive Analytics:

- Order volumes are expected to remain highest in **November and December**, driven by seasonal and **festive demand**.
- **September and October** will likely sustain moderate traction, continuing as secondary contributors.
- **May** and similar off-season months are projected to stay weaker unless supported by targeted campaigns.
- Overall, peak months will continue to dominate, with **60–70%** of annual orders concentrated in festive seasons unless demand is diversified.

4. Prescriptive Analytics:

- Focus on **November and December** with strong inventory planning, festive promotions, and bundled offers to maximize peak demand.
 - Strengthen **September and October** through targeted campaigns, loyalty programs, or early seasonal launches to boost mid-level traction.
 - Introduce special promotions in **May and other weaker months** to spread demand more evenly across the year.
 - A balanced strategy will **reduce reliance on peak months** while ensuring consistent order flow and long-term profitability.
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8.6: TOTAL ORDER BY CITY:



The following KPIs were identified from the column chart analysis

1. Descriptive Analytics:

- **New York** stands out with nearly 1000 orders, making it the clear leader (Specific).
- The top **three cities together** contribute over **40%** of total demand, showing measurable concentration (**Measurable**).
- Mid-tier cities like **Seattle and Houston** present achievable growth opportunities with targeted strategies (**Achievable & Relevant**).
- Focused interventions in **low-order cities** should be time-bound within the next year to balance performance (**Time-bound**).

2. Diagnostic Analytics:

- **New York** has the highest orders because of strong demand and better market presence.
- **Los Angeles and Philadelphia** are strong but may need more promotions to catch up.
- **Seattle, Houston, Chicago** show moderate performance, likely due to partial awareness or logistics limits.
- **Columbia, Jackson, Richmond** are very low, pointing to weak visibility or customer base issues.

3. Predictive Analytics:

- **New York** is expected to continue leading, with orders likely crossing 1000+ if current growth trends hold.
- **Los Angeles and Philadelphia** may steadily rise and could narrow the gap with New York if demand accelerates.
- **Mid-tier cities (Seattle, Houston, Chicago)** are projected to grow moderately, reaching **600–700 orders** in the next cycle with targeted campaigns.
- Low-performing cities (**Columbia, Jackson, Richmond**) will remain below **300** unless strong marketing or distribution strategies are introduced.

4. Prescriptive Analytics:

- **New York** should be sustained with loyalty programs to secure its lead.
 - **Los Angeles and Philadelphia** need promotions and faster delivery to boost demand.
 - **Seattle, Houston, and Chicago** can grow through local marketing and stronger logistics.
 - **Columbia, Jackson, and Richmond** require awareness campaigns and better distribution to uplift performance.
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9. CONCLUSION:

The business is profitable with **\$93.4K** profit and steady growth across **2021–2024**. **New York and the Pacific** region dominate performance, while **Milk Chocolate** leads product sales. Seasonal demand peaks in **November and December**, and **Standard** shipping is the preferred mode. The conclusion: sustain top performers, strengthen mid-tier regions, and uplift weaker cities for balanced expansion.
